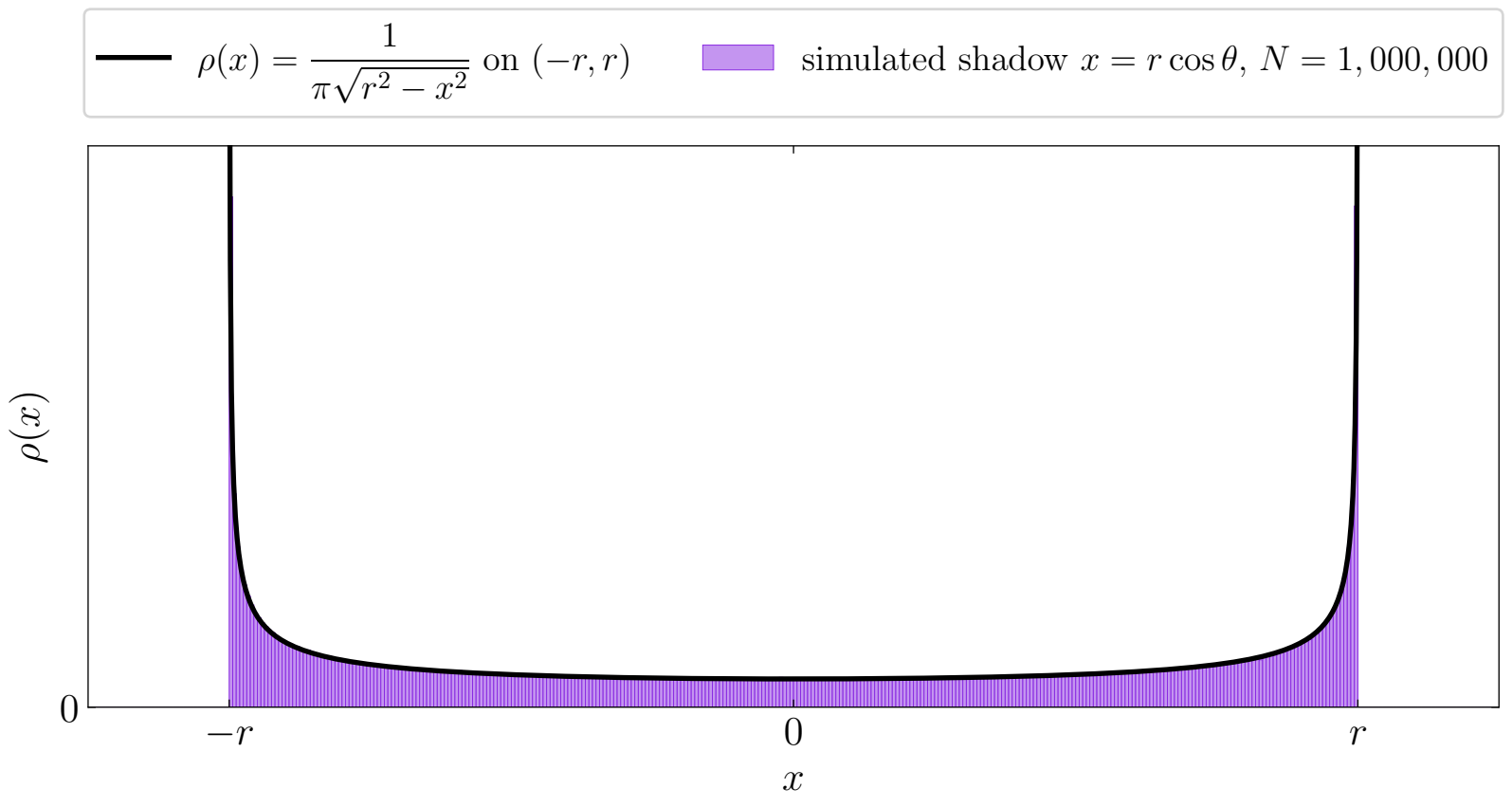
$\rho(x) = \dfrac{1}{\pi\sqrt{r^2 - x^2}}$ on $(-r, r)$
simulated shadow $x = r\cos\theta$, $N = 1,000,000$
$\rho(x)$
0
$-r$
0
r
x